Lesson notes · for you

Wrong examples, wrong answers

AI security · Primary school · Lesson 2

Time
20 min

Materials
Ten small cards with simple drawings of two kinds of fruit (or real leaves of two kinds); A pencil to write a label on each card

Learning goals

- Explain how AI learns from examples with labels.
- Show what wrong labels do to an AI.
- Find and fix the odd ones out.

Activity: Activity: Teach the sorter

1. Make (5 min): Draw ten small cards: five of one fruit and five of another, such as mango and lemon. Write the right label on each card.
2. Teach (5 min): Your friend is the “computer”. Show them all ten cards with their labels. Then hold up five new drawings and ask them to sort each one.
3. Trick (5 min): Make one new card with the wrong label, for example a big mango labelled “lemon”. Teach your friend again with this card mixed in, then sort new drawings again. Does your friend make more mistakes?
4. Find (5 min): Lay all the cards out together. Which ones look odd next to the others with the same label? Fix those labels, and talk about how you found them.

No cards? Use real leaves of two kinds, or stones of two colours.

Discussion

- Why can't a computer tell when a label is wrong?
- Why might someone add wrong examples on purpose?
- How can you help keep examples right?

Check yourself

- Explain in one sentence why wrong labels make wrong answers.

Try it this week
Play “Teach the sorter” at school or at home: one friend is the computer and learns from picture cards with labels. Sneak in one wrong label, and see what the “computer” gets wrong.

Worksheet

Wrong examples, wrong answers

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Name _____ Class _____ Date _____

1 **Your class is teaching a computer to sort leaves. A friend labels a few wrongly, as a joke. What should you do?**

- ◯ Nothing: it's only a few leaves
- ◯ Fix the labels and tell your teacher
- ◯ Add more wrong labels for fun

2 **Put in order: how a computer learns to sort fruit.**

- ☐ It sorts new fruit by itself
- ☐ The computer finds what's alike
- ☐ Give each picture a label
- ☐ Collect pictures of fruit

3 **A computer can always tell when one of its labels is wrong.**

- ◯ True
- ◯ False

4 **Who could put wrong labels into the examples? Choose all that apply.**

- ☐ Someone in a hurry
- ☐ Someone playing a trick on purpose
- ☐ A careful helper who checks twice
- ☐ A friend labelling for a joke

5 **Match each problem to its fix.**

A label written in a hurry	• • • •	Ask where they came from
A strange example among the others	• • • •	Look closer, then fix or remove it
Examples from someone you don't know	• • • •	Check it twice

✎ **Have you ever learned something wrong because someone told you wrong? How did you find out?**